Paper SCBU — Substrate-Cosmology Boundary Unification

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May 2026

Preamble — What This Paper Does NOT Claim

1. This paper does **not** claim that the MOND acceleration scale a_0 and the ED-SC canonical operating point $\xi_{\text{canonical}}$ are numerically identical or equal at substrate level; they are scales of different physical dimensions (acceleration vs. length).
2. It does **not** claim a substrate-derivation of the numerical value of $\xi_{\text{canonical}}$ from H_0 analogous to Paper_029's derivation of a_0; that derivation is OPEN.
3. It does **not** claim that the unified substrate-cosmology boundary structure proposed here is the **only** possible cross-arc unification of MOND and ED-SC content.
4. It does **not** introduce new substrate primitives; only one paper-specific postulate (P-Substrate-Cosmology-Unified) is introduced.
5. It does **not** override or strengthen any constituent-paper verdict; it composes Paper_028, Paper_029, Paper_096, and Paper_097 into a single substrate-cosmology framing.
6. It does **not** claim novel empirical predictions; cross-arc structural identification is the deliverable, not new phenomenology.
7. "Substrate-cosmology boundary" = the substrate-level surface at $R_H = c/H_0$ (Paper_028) beyond which cross-locus substrate influence cannot reach coherently. Both MOND and ED-SC content inherit projection structure from this boundary.

Abstract

The MOND acceleration scale $a_0 = cH_0/(2\pi)$ (Paper_029) and the ED-SC canonical operating point $\xi_{\text{canonical}} = 1.7575$ lu (Paper_096) are anchored to substrate-cosmology boundary structure but have no current unifying paper in the corpus. This synthesis paper claims that both scales are projections of a single substrate-cosmology boundary at $R_H = c/H_0$ (Paper_028): the MOND projection arises via azimuthal-Fourier on the residual $SO(2)$ symmetry of an accelerating chain (Paper_029 §5.1 mechanism); the ED-SC projection arises via cross-scale-invariance fixed-point matching on the kernel hierarchy (Paper_096 mechanism). The two projections share the same substrate-cosmology origin and are FORCED to be structurally related under a single new postulate (P-Substrate-Cosmology-Unified), while remaining numerically distinct because they project the same boundary onto different observables (acceleration vs. substrate-length-unit). The synthesis prepares the corpus for ED-SC 4.x extensions to BH, MOND/galactic, Q-Compute, and soft-matter scales. The structural form is FORM-FORCED; the substrate-derivation of $\xi_{\text{canonical}}$ from H_0 remains OPEN.

1 Introduction

1.1 The cross-arc gap

The ED corpus contains two independent canon-internal scale anchors of substrate-cosmological origin:

- **Paper_029** derives $a_0 = cH_0/(2\pi)$ as the substrate-level MOND transition acceleration. The factor $1/(2\pi)$ arises from azimuthal-Fourier-mode normalization on the residual $SO(2)$

symmetry of an accelerating chain projecting the cosmic decoupling surface (Paper_028).
- **Paper_096** establishes the ED-SC cross-scale invariance with canon-internal canonical operating point $\xi_{\text{canonical}} = 1.7575$ lu. This value is inherited from the ED-SC 3.3.x arc closure.

A cross-arc audit identified that no current paper unifies these two scales as projections of a common substrate-cosmology boundary structure. Both are anchored at $R_H = c/H_0$ in upstream content (Paper_028), but the substrate-mechanism linking them across the MOND-arc and Wedges-arc has not been formalized. This paper supplies that formalization at verdict tier M3 per Paper_095 methodology.

1.2 Why unification matters

The substrate-cosmology boundary $R_H = c/H_0$ is a load-bearing structural object in two arcs simultaneously:

- **MOND arc:** R_H is the cosmic decoupling surface (Paper_028) supplying the geometric and dipole-projection structure that produces a_0.
- **Wedges arc:** R_H is the implicit far-IR endpoint of the three-regime RG flow (Paper_097), beyond which cross-scale invariance ceases to apply at substrate level.

If both arcs invoke the same substrate-cosmology boundary, then their canon-internal scale anchors should be structurally related. The current corpus has the boundary referenced in both arcs but the **two anchors are not currently shown to inherit from it jointly**. This is a substrate-level structural gap, not an empirical gap.

Joint inheritance is a substrate-level structural claim, not a citation-hygiene issue — it requires its own postulate and a derivation chain composing Paper_028 + Paper_029 + Paper_096 + Paper_097.

1.3 How this paper fits in the corpus

This paper sits at the intersection of the gravity-arc and wedges-arc, analogous to:

- **Paper_062** (cross-domain echo BH $\leftrightarrow$ Q-COMPUTE via shared V5 mechanism) — methodological template.
- **Paper_071** (ER=EPR-class structural echo Entanglement $\leftrightarrow$ BH via shared V5 mechanism) — methodological template.

Both Paper_062 and Paper_071 introduce single shared-mechanism postulates linking arcs that otherwise have independent derivation chains. This paper follows the same template: a single shared-mechanism postulate (P-Substrate-Cosmology-Unified) linking MOND arc and Wedges/RG arc via Paper_028's substrate-cosmology boundary.

The structural-echo verdict in §6 is **A→position** — same substrate boundary, two projection mechanisms — not a new derivation of either a_0 or $\xi_{\text{canonical}}$ from first principles.

2 Primitive Inputs and Paper-Specific Postulates

2.1 Primitives invoked (per Paper_087)

- **P03 (Channel + locus indexing; spatial homogeneity)** — supplies substrate-level azimuthal and translation symmetry against which dipole-projection and cross-scale-invariance fixed-points are defined.
- **P08 (Substrate scale ℓ_{ED})** — supplies the substrate length scale entering $\xi_{\text{canonical}}$ via Paper_096 inheritance.

- **P11 (Commitment-irreversibility)** — supplies the directional substrate evolution underlying the cosmic decoupling surface (Paper_028) and the IR-regime cutoff (Paper_097).
- **P13 (Time homogeneity)** — supplies H_0 as the cosmological rate parameter against which R_H is defined.

2.2 Upstream dependencies (I-rows)

- **I-028:** Cosmic decoupling surface $R_H = c/H_0$ (Paper_028). [**LOAD-BEARING — shared boundary source**]
- **I-029:** $a_0 = cH_0/(2\pi)$ via dipole-projection on residual $SO(2)$ symmetry (Paper_029). [**LOAD-BEARING — MOND projection mechanism**]
- **I-091:** Kernel cascade across scales (Paper_091). [**LOAD-BEARING — Wedges-arc scale-hierarchy framework**]
- **I-096:** ED-SC cross-scale invariance with $\xi_{\text{canonical}}$ canonical operating point (Paper_096). [**LOAD-BEARING — ED-SC projection mechanism**]
- **I-097:** Three-regime RG with canon-internal 0.6 transition exponent (Paper_097). [**LOAD-BEARING — IR-regime endpoint**]
- **I-037:** a_0 cosmological-rate invariance under local rescaling (Paper_037 P-H0-Cosmological-Invariant). [**CONTEXT**]
- **I-062:** Cross-domain echo via shared substrate mechanism (Paper_062). [**CONTEXT — methodological template**]
- **I-071:** ER=EPR-class structural echo (Paper_071). [**CONTEXT — methodological template**]
- **I-095:** Form-FORCED / value-INHERITED methodology with three-tier verdict classification (Paper_095). [**METHODOLOGY**]

2.3 Paper-specific postulates

P-Substrate-Cosmology-Unified: The MOND acceleration scale $a_0 = cH_0/(2\pi)$ (Paper_029) and the ED-SC canonical operating point $\xi_{\text{canonical}}$ (Paper_096) are two projections of a single substrate-cosmology boundary structure anchored at $R_H = c/H_0$ (Paper_028). The MOND projection is azimuthal-Fourier on residual $SO(2)$ symmetry of an accelerating chain (Paper_029 §5.1 mechanism); the ED-SC projection is cross-scale-invariance fixed-point on the kernel hierarchy (Paper_096 mechanism). The two projections share the same substrate-cosmology origin, are FORCED to be structurally related (via the shared boundary), and are constrained to scale jointly under H_0 variation. Numerical values of a_0 and $\xi_{\text{canonical}}$ remain INHERITED — the postulate fixes the structural relationship, not the numerical magnitudes.

This is the single new postulate introduced by this paper. It follows the cross-domain-echo template of Paper_062 (P-V5-Shared-Mechanism) and Paper_071 (P-V5-Shared-EntanglementGravity): a single load-bearing postulate that claims a shared substrate object underlies otherwise-independent arc derivations.

§2.5 Load-Bearing Step Audit Table

#	Step	Label	Notes
1	Substrate-cosmology boundary $R_H = c/H_0$	I	Paper_028.
2	$a_0 = cH_0/(2\pi)$ via dipole-projection	I	Paper_029.
3	ED-SC cross-scale invariance with $\xi_{\text{canonical}}$	I	Paper_096.
4	Three-regime RG with 0.6 transition exponent	I	Paper_097.
5	Memory-kernel cascade across scales	I	Paper_091.
6	Both a_0 and $\xi_{\text{canonical}}$ anchored to R_H	P-Substrate-Cosmology-Unified	Postulate.
7	MOND projection: azimuthal-Fourier on residual $SO(2)$	D-via-I	Paper_029 §5.1 mechanism.
8	ED-SC projection: cross-scale-invariance fixed-point	D-via-I	Paper_096 P-S1-Invariant mechanism.
9	Two projections share substrate-cosmology origin	D	Composition of steps 6–8.
10	Joint scaling under H_0 variation	D	Composition of step 9 + I-037 invariance.
11	Numerical values of a_0 and $\xi_{\text{canonical}}$	I	Empirical via Paper_029 + canon-internal via Paper_096.
12	Substrate-derivation of $\xi_{\text{canonical}}$ from H_0	OPEN	Not claimed.
13	Cross-arc echo structural-positive verdict	A→position	Composite.

Verdict tier per Paper_095: **M3 — form-FORCED + value-INHERITED.** Potential upgrade to M2 (Intermediate Path C) if step 12 closes — i.e., if a substrate-derivation of $\xi_{\text{canonical}}$ from H_0 is supplied.

3 The Unification

3.1 The shared substrate-cosmology boundary

Paper_028 establishes the cosmic decoupling surface at $R_H = c/H_0$ as the substrate-level boundary beyond which cross-locus substrate influence cannot reach a local chain coherently. This boundary is a structural property of the substrate participation graph under P11 commitment-irreversibility + P13 time homogeneity + the kernel-propagation rate c. The substrate-level identity:

$$R_H = c/H_0$$

is the canonical substrate-cosmology boundary. Two distinct projections of this boundary appear in the corpus.

3.2 The MOND projection (Paper_029 mechanism)

By Paper_029, an accelerating chain breaks the full $SO(3)$ rotational symmetry of substrate-graph adjacency to the residual $SO(2)$ symmetry about the acceleration axis. The substrate-graph response to the cosmic decoupling surface projects this content via azimuthal-Fourier-mode normalization. The leading anisotropic mode is the dipole ($l = 1$), and the $|m| = 1$ Fourier projection on the residual $SO(2)$ carries the geometric factor $1/(2\pi)$:

$$a_0 = \frac{cH_0}{2\pi}.$$

The factor $1/(2\pi)$ is FORM-FORCED by the azimuthal-Fourier normalization on the circle of period 2π. The numerical $a_0 \approx 1.08 \times 10^{-10}\,\text{m/s}^2$ at $H_0 \approx 70$ km/s/Mpc inherits Hubble-rate observational content.

3.3 The ED-SC projection (Paper_096 mechanism)

By Paper_096, the substrate kernel hierarchy (Paper_092 three-index classification) admits a cross-scale invariance under scale transformations within the hydrodynamic-window-extended regime. The substrate-level S1 kernel-content is invariant under rescaling; the S2 ensemble-only content is regime-specific. The cross-scale-invariance fixed point is the canonical operating point:

$$\xi_{\text{canonical}} = 1.7575\,\text{lu}.$$

The numerical value is canon-internal — inherited from ED-SC 3.3.x arc closure. At substrate level, $\xi_{\text{canonical}}$ is anchored to the substrate-cosmology boundary $R_H = c/H_0$ via the IR-regime endpoint of the three-regime RG flow (Paper_097): beyond R_H, cross-scale invariance ceases to apply at substrate level, and $\xi_{\text{canonical}}$ is the canonical operating point at which the invariance is maximally saturated within the IR regime.

3.4 Joint inheritance from R_H

By P-Substrate-Cosmology-Unified, both a_0 and $\xi_{\text{canonical}}$ inherit structurally from the same substrate-cosmology boundary $R_H = c/H_0$. The two projections are mechanistically distinct:

- **MOND projection** projects R_H onto **accelerating-chain content** via residual $SO(2)$ Fourier.
- **ED-SC projection** projects R_H onto **substrate-kernel-hierarchy content** via cross-scale-invariance fixed point.

But they share the same substrate-cosmology origin: R_H is the load-bearing structural object in both cases. The structural relationship between the two projections is FORCED by joint inheritance from a single boundary.

3.5 Joint scaling under H_0 variation

A direct consequence of joint inheritance: under variation of H_0 (e.g., Hubble-tension resolution at any value in the empirical range), both a_0 and $\xi_{\text{canonical}}$ must scale jointly. Specifically:

- a_0 scales linearly with H_0: $a_0 = cH_0/(2\pi)$ — direct.
- $\xi_{\text{canonical}}$ scales with H_0 through its anchoring to $R_H = c/H_0$ — indirect.

The exact functional form of the $\xi_{\text{canonical}}(H_0)$ scaling is not derived in this paper (OPEN at step 12 of the audit table); only the structural claim that the two scales co-vary under H_0 is FORCED. If empirical H_0 measurements pin down a value in the Hubble-tension range $[60, 80]$ km/s/Mpc, a_0 is predicted in the range $[0.93, 1.24] \times 10^{-10}$ m/s^2 and $\xi_{\text{canonical}}$ should scale correspondingly. Independent variation of one without the other would refute P-Substrate-Cosmology-Unified.

3.6 What this is NOT

The unification is **structural**, not operational:

- It does **not** claim that a_0 and $\xi_{\text{canonical}}$ are numerically equal — they have different physical dimensions.
- It does **not** claim that the MOND mechanism and the ED-SC mechanism are operationally equivalent — they project the same boundary via different operations (azimuthal-Fourier vs. cross-scale-invariance fixed-point).
- It does **not** supply a first-principles substrate-derivation of $\xi_{\text{canonical}}$ from H_0 — only the inheritance pattern is forced; the substrate-derivation analogous to Paper_029's a_0 derivation is OPEN.

This is **A→position** at the composite-verdict level (audit-table row 13), following the cross-domain-echo template of Paper_062 §3.4 and Paper_071 §3.4.

4 Distinguishing Structural vs. Numerical Content

Structural (D-via-I):

- Joint inheritance of a_0 and $\xi_{\text{canonical}}$ from R_H.
- Mechanism-distinctness of the two projections (azimuthal-Fourier vs. cross-scale-invariance fixed-point).
- Joint scaling of a_0 and $\xi_{\text{canonical}}$ under H_0 variation.

Inherited (I):

- Numerical value of a_0 (via Paper_029).
- Numerical value of $\xi_{\text{canonical}}$ (canon-internal via Paper_096).
- Numerical value of H_0 (empirical cosmological observation).
- Numerical value of 0.6 transition exponent (canon-internal via Paper_097).

OPEN:

- Substrate-derivation of $\xi_{\text{canonical}}$ from H_0 analogous to Paper_029's derivation of a_0.
- Quantitative functional form of $\xi_{\text{canonical}}(H_0)$ scaling.
- Extension to additional substrate-cosmology-anchored scales (BH horizon $r_H = 2GM/c^2$, Q-Compute $\mathcal{M}_{\text{crit}}$, soft-matter L_{flow}).

5 Falsification Criteria

- **F1:** Empirical evidence that a_0 and $\xi_{\text{canonical}}$ scale **independently** under H_0 variation — refutes P-Substrate-Cosmology-Unified joint-inheritance claim. This is the sharpest falsifier: if Hubble-tension resolution moves H_0 to a specific value and only one of the two anchors tracks the change, the unified-source claim fails.
- **F2:** Independent substrate-derivation of $\xi_{\text{canonical}}$ that does NOT invoke H_0 or substrate-cosmology — refutes the boundary-unification mechanism. If $\xi_{\text{canonical}}$ is shown to derive from a purely substrate-internal mechanism (e.g., V1 kernel-width self-consistency at substrate scale alone), then the cross-arc unification dissolves.
- **F3:** Refutation of Paper_028 (cosmic decoupling surface), Paper_029 (a_0 formula), or Paper_096 (cross-scale invariance) propagates upward to refute this paper. The synthesis collapses if any single upstream pillar is refuted.
- **F4:** Empirical or substrate evidence that a third or fourth canon-internal scale anchor (e.g., BH horizon r_H, Q-Compute $\mathcal{M}_{\text{crit}}$, NS-Q operating point) is **not** anchored to $R_H = c/H_0$ — would force re-examination of the substrate-cosmology-boundary framework's claimed cross-arc reach.
- **F5:** Discovery of a fifth independent projection mechanism on R_H producing a third distinct cosmologically-anchored scale not currently in the corpus — would extend the framework rather than refute it.

6 Position Statement

The MOND acceleration scale $a_0 = cH_0/(2\pi)$ (Paper_029) and the ED-SC canonical operating point $\xi_{\text{canonical}} = 1.7575$ lu (Paper_096) are **FORCED to share** a single substrate-cosmology boundary origin at $R_H = c/H_0$ (Paper_028) under P-Substrate-Cosmology-Unified. The two projections are mechanistically distinct (azimuthal-Fourier on residual $SO(2)$ for MOND; cross-scale-invariance fixed-point on kernel hierarchy for ED-SC) but inherit structurally from the same boundary. Joint scaling under H_0 variation is FORCED; numerical values remain INHERITED.

The result is at verdict tier M3 per Paper_095 methodology. Upgrade to M2 (Intermediate Path C) requires substrate-derivation of $\xi_{\text{canonical}}$ from H_0 analogous to Paper_029's derivation of a_0 — OPEN.

This synthesis closes the cross-arc harmonization gap between the MOND arc and the Wedges arc. With this paper, **substrate-cosmology boundary structure is globally consistent across MOND arc, Wedges arc, and the cross-scale-invariance framework**. The result joins Paper_062 (BH $\leftrightarrow$ Q-COMPUTE cross-domain echo) and Paper_071 (Entanglement $\leftrightarrow$ BH ER=EPR-class echo) as the third corpus-level cross-arc synthesis paper at structural-echo verdict.